Date:

# Section no 2
# Probability

## Uncertainity Due to Randomness:

Uncertainity: outcome of an experiment is not known in advance.

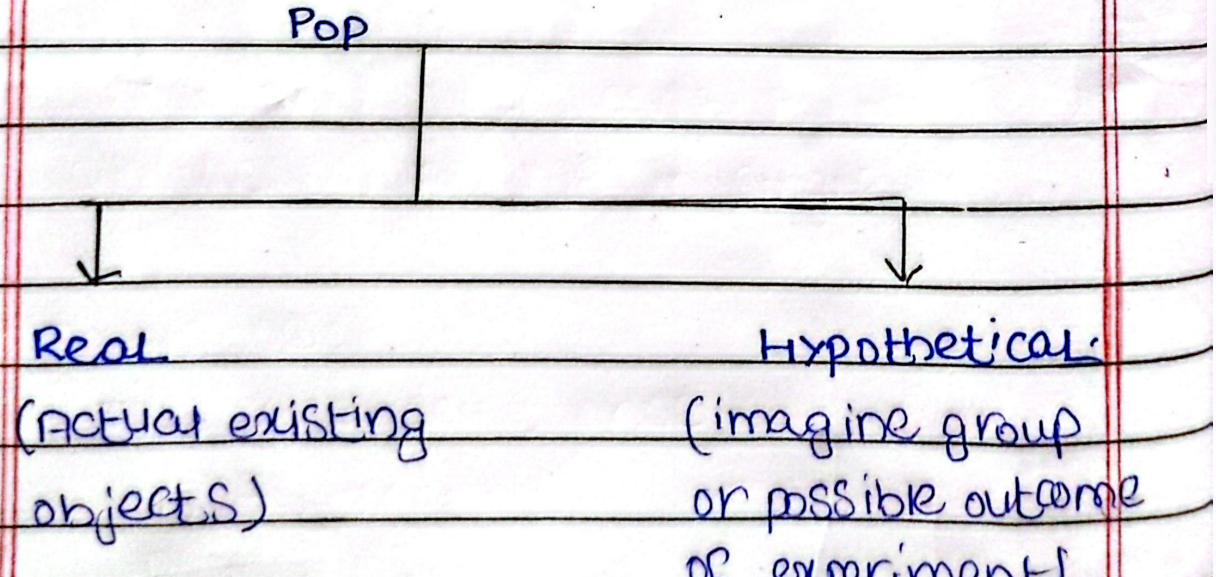
marks = f(Std Hrs, Travel Hrs)

we can control — we can manage

But:

accident: these are random events. they may effect results.

Probability: study that studies the chance of an event happening.



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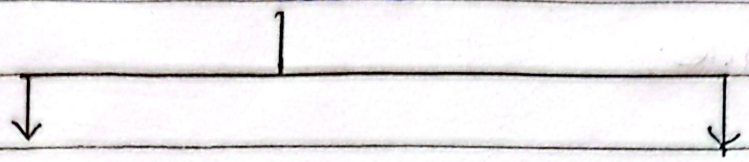
Result



Predictable

If predicable then Deterministic.

Unpredictable

If unpredictable event when repeated Large no of time under same condition then Random.

**Event:** any set of outcome from sample space.

**Sample space:** set of all possible outcomes of experiment.

**outcome:** single, uniqe possible result of experiment eg H or t

**Trial:** Performing an experiment once e.g Rolling die one time.

**Null Event:** impossible event e.g Getting 5 & 6 on single Die

**Sure Event:** event that must happen e.g two coins are tossed

$S = \{HH, HT, TH, TT\}$ → sure event

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Simple Event: that contain one outcome.

Compound Event: contain two or more outcome.

$S = \{HH, HT, TH, TT\}$

$\emptyset$ → Null event

$\{HH\}, \{HT\}, \{TH\}, \{TT\}$ → Simple event

$\{HH, HT\}, \{HH, TH\}, \{HH, TT\}$
$\{HT, TH\}, \{HT, TT\}$
$\{TH, TT\}$
$\{HH, HT, TH\}, \{HH, TH, TT\}$
$\{HT, TH, TT\}$
→ Compound Event

$\{HH, HT, TH, TT\}$ → Sure event and Compound.

contain all possible outcome & must happen.

Mutually Exclusive:

Two events that can't happen at the same time in single trial.

Example:

Tossing a coine once.

$A = \{H\}$

$B = \{T\}$

Date: _____ Day: _____

we can't get both Head and Tail in one toss.

$$A \cap B = \emptyset$$

**Exhaustive Event:**

Two or more events are exhaustive if together they cover all possible outcomes of an experiment.

Example: Rolling Die

$A = \{2, 4, 6\}$

$B = \{1, 3, 5\}$

Together

$$A \cup B = \{1, 2, 3, 4, 5, 6\}.$$

Now A and B are exhaustive coz they cover every possible outcome.

**Equally Likely Event:**

Events that have same/Fair chance of occuring.

Example: Tossing a fair coin both H and T have equally chance:

Date: _____ Day: _____

# Probability.

## Objective

- Measure by some well define methods
- Prior knowledge matters
- Based on Logic & Reasoning.

### Classical Definition

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(\text{event}) = \frac{\text{No of Favorable Outcomes}}{\text{No of Possible Outcomes}}$$

### Relative Freq.

### Axiomatic Definition.

## Subjective:

Based on someone personal Judgement or belief

**Permutation:-**

is an arrangements of objects in which order matters

**Example:**

Arrange A & B

Date: _____ Day: _____

- Possible Arrangements : AB, BA
- Here AB and BA are different.

Formula:

$$^nP_r = \frac{n!}{(n-r)!}$$

**Combination:**

Selection of object in which order does't matter.

Example:

- Select 2 students A and B
- Possible selection : AB
- AB and BA represent same selection.

Formula:

$$^nC_r = \frac{n!}{r!\,(n-r)!}$$

## Addition Law For Probability:

Find Probab. that atleast one of two events (A or B) will occur.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- $P(A \cup B)$: Prob that either event A or Event B occurs
- $P(A)$: Prob of A
- $P(B)$: Prob of B
- $P(A \cap B)$: Prob of A and B occuring at same time.

But if two event can't happen at same time.

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ → becomes 0

Date: $P(A \cup B) = P(A) + P(B)$ Day:

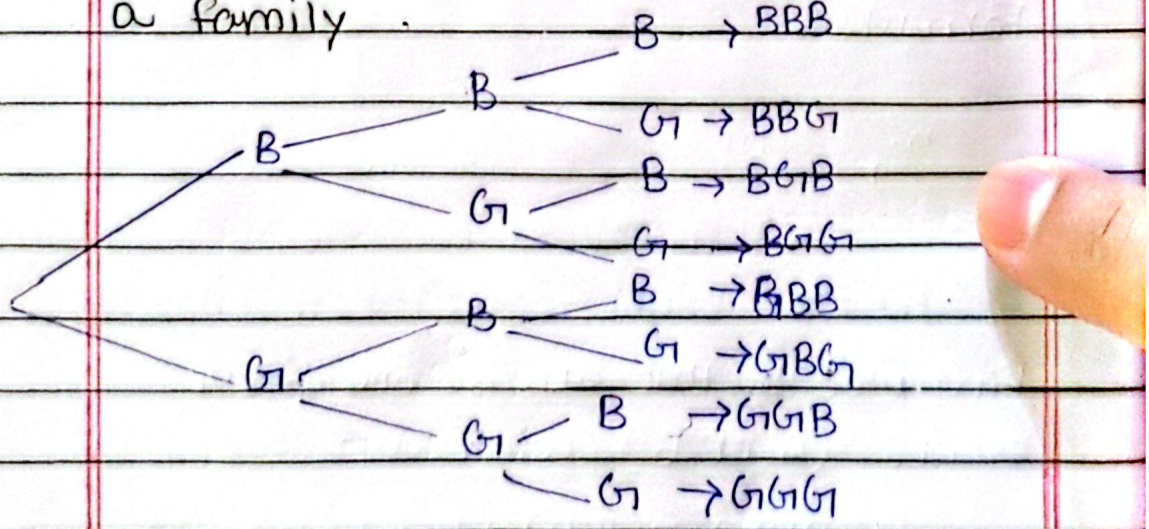
**Example:** Bluman Pg # (183)

Find Sample space For rolling two Dice.

| | Die 2 | | | | | |
|---|---|---|---|---|---|---|
| Die 1 | 1 | 2 | 3 | 4 | 5 | 6 |
| 1 | (1,1) | (1,2) | (1,3) | (1,4) | (1,5) | (1,6) |
| 2 | (2,1) | (2,2) | (2,3) | (2,4) | (2,5) | (2,6) |
| 3 | (3,1) | (3,2) | (3,3) | (3,4) | (3,5) | (3,6) |
| 4 | (4,1) | (4,2) | (4,3) | (4,4) | (4,5) | (4,6) |
| 5 | (5,1) | (5,2) | (5,3) | (5,4) | (5,5) | (5,6) |
| 6 | (6,1) | (6,2) | (6,3) | (6,4) | (6,5) | (6,6) |

**Example:** Bluman Pg # (185)

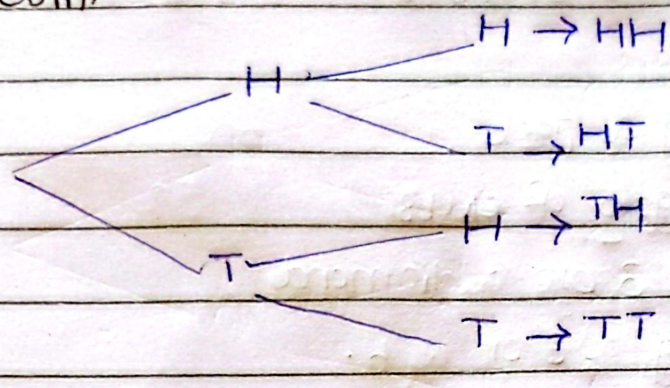
Use a tree diagram to find sample space for gender of three children in a family.



Date: _____ Day: _____

**Example:**

Find Sample space For Tossing Two coin.




**Example: 4-5** Bluman Pg # 187

Find a Probability of getting black 10 when drawing from a deck.

one black 10 is in club = 1

one black 10 is in Spade = 2

Let P(black 10)

$$P(\text{black 10}) = \frac{2}{52} = 0.038$$

**Example: 4-6** Bluman Pg # 187.

If family has three children find two of three children are girls.

Let event E: (two of childre three are girl)

$$P(E) = \frac{3}{8} = 0.375$$

Date: _____ Day: _____

## Example: 4-7 (Burman Pg # 187)

Card is drawn from an ordinary deck;
Find these probabilities.

a.) of getting a Jack
b) of getting 6 of clubs
c) of getting 3 or a diamond
d) of getting a 3 or a 6.

### a

Given Data :-

Total card = 52 = n(S)
Select = 1
Total Jack = 4

Let event E: ( of getting a Jack)

$$P(E) = \frac{n(E)}{n(S)} = \frac{\binom{4}{1}\binom{48}{0}}{\binom{52}{1}} = \frac{(4)(1)}{52}$$

$$P(E) = 0.077$$

$$\boxed{P(E) = 7.7\%}$$

Date: _____ Day: _____

## b-

Data:

Total Card = 52

Select = 1

6 of Club = 1

Let event : G1 (of getting 6 of clubs)

$$P(G_1) = \frac{n(G_1)}{n(S)} = \frac{\binom{1}{1}\binom{51}{0}}{\binom{52}{1}} = \frac{(1)(1)}{52}$$

$P(G_1) = 0.019$

$\boxed{P(G_1) = 1.9\%}$

## c-

of Getting 3 or a Diamond:

Data

total Card = 52

Select = 1

Total 3 Card = 4

Total Diamond = 13

getting 3 of Diamond = ?1

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

P(A) : Let event A: (getting 3)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{4}{1}\binom{48}{0}}{\binom{52}{1}} = \frac{(4)(1)}{52}$$

Date: Day:

$P(A) = \frac{4}{52} = 0.077$

Let event B: (getting a Diamond)

$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{13}{1}\binom{39}{0}}{\binom{52}{1}} = \frac{(13)(1)}{52}$

$P(B) = \frac{13}{52} = 0.25$

Let event $(A \cap B)$: (getting 3 of Diamond)

$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{\binom{1}{1}\binom{51}{0}}{\binom{52}{1}} = \frac{(1)(1)}{52}$

$P(A \cap B) = \frac{1}{52} = 0.019$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = 0.077 + 0.25 - 0.019$

$P(A \cup B) = 0.308$

$\boxed{P(A \cup B) = 30.8\ \%}$

d-

of getting 3 or a 6.

Total card = 52

Select = 1

3 card = 4

6 card = 4

3 and 6 at same time = 0

Date: _____ Day: _____

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = P(A) + P(B) - 0$

$P(A \cup B) = P(A) + P(B)$

Let event A: (of getting 3)

calculated is C part

$P(A) = 0.0777$ or $\frac{4}{52}$

Let event B: (of getting 6)

$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{4}{1}\binom{48}{0}}{52} = \frac{(4)(1)}{52}$

$P(B) = \frac{4}{52} = 0.0777$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = \frac{4}{52} + \frac{4}{52} - 0$

$P(A \cup B) = \frac{8}{52} = \frac{2}{13} = 0.154$

$\boxed{P(A \cup B) = 15.4\ \%}$

Date: Day:

**Example:**

A committee of 3 persons is to be selected randomly from a group of 5 men & 3 women. Find

a) Probability that committee consist of 1 men & women.

b) Prob that committee consist of all women.

Given Data:

Total members = 8 = n(S)

Select = 3 Men = 5

Women = 3

**a:**

Let event A: (1 men & Women)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{5}{1}\binom{3}{2}}{\binom{8}{3}} = \frac{(5)(3)}{56}$$

$$P(A) = \frac{15}{56} = 0.268$$

$$P(A) = 26.8\ \%$$

Date: _____ Day: _____

## b-

Let event B: (ALL women)

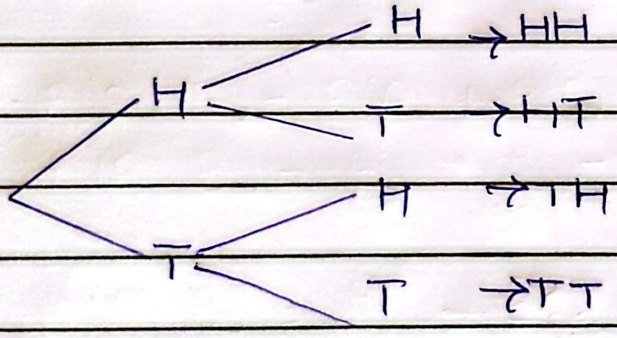
$$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{3}{3}\binom{5}{0}}{\binom{8}{3}} = \frac{(1)(1)}{56}$$

$$P(A) = \frac{1}{56} = 0.018$$

$$\boxed{P(A) = 1.8\ \%}$$

**Example:**

Two Fair coins or tossed what is prob of getting 1 Head & 1 Tail.




Let A: (Prob of 1 Head & 1 Tail)

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = 0.5$$

$$\boxed{P(A) = 50\ \%}$$

Date: _____ Day: _____

## Example:

A coin is toss twice what is prob that atLeast one Head occur-

$$S = \{HH, HT, TH, TT\}.$$

Let event A: (atLeast one head)

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{4} = 0.75$$

$$\boxed{P(A) = 75\%}$$

## Example:

Q:

Data:-

Total = 6 , Select = 2

defective = 2 , non-defective = 4

(1) P(Atleat one is defective)

Let event : D (Atleat one is defective)

$$P(D) = \frac{n(D)}{n(S)} = \frac{\binom{2}{1}\binom{4}{1} + \binom{2}{2}\binom{4}{0}}{\binom{6}{2}}$$

$$P(D) = \frac{(2)(4)+(1)(1)}{15} = \frac{9}{15} = 0.6$$

(2)

Let event : E (both are defective)

$$P(E) = \frac{n(E)}{n(S)} = \frac{\binom{2}{2}\binom{4}{0}}{\binom{6}{2}} = \frac{(1)(1)}{15}$$

$$P(E) = 0.06$$

b-

Data:

Total = 6, Select = 4

defective = 4, non defective = 2

(1) P(atleast one is defective)

Let event D: (atleast one is defe)

$$P(D) = \frac{n(D)}{n(S)} = \frac{\binom{4}{1}\binom{2}{1} + \binom{4}{2}\binom{2}{0}}{\binom{6}{4}}$$

$$P(D) = \frac{(4)(2) + (6)(1)}{15} = \frac{8+6}{15} = 0.93$$

(2) P(Both are defective)

Let event E: (Both are defe)

$$P(E) = \frac{n(E)}{n(S)} = \frac{\binom{4}{2}\binom{2}{0}}{\binom{6}{4}} = \frac{(6)(1)}{15} = 0.4$$

S M T W T F S DATE:

# Question: 14

## Q5

Data:-

Total = 10 Select = 3

Yellow = 3, Black = 7

(1) P(exactly one Yellow Fish)

Let event Y: (exactly one Yellow)

$$P(Y) = \frac{n(Y)}{n(S)} = \frac{\binom{3}{1}\binom{7}{2}}{\binom{10}{3}} = \frac{(3)(21)}{120} = \frac{63}{120}$$

$$P(Y) = \frac{21}{40} = 0.525$$

## b-

total = 10, Select = 3

Yellow = 3, Seven = 7

P(atmost one Yellow Fish)

Let event D: (atmost one Yellow)

$$P(D) = \frac{n(D)}{n(S)} = \frac{\binom{3}{1}\binom{7}{2} + \binom{3}{0}\binom{7}{3}}{\binom{10}{3}} =$$

$$P(D) = \frac{(3)(21) + (1)(35)}{120} = \frac{98}{120}$$

$P(D) = 0.817$

**C-**

P (atLeast one yellow Fish)

Let event E: (atleast one yellow)

$$P(E) = \frac{n(E)}{n(S)} = \frac{\binom{3}{1}\binom{7}{2} + \binom{3}{2}\binom{7}{1} + \binom{3}{3}\binom{7}{0}}{\binom{10}{3}}$$

$$P(E) = \frac{(3)(21) + (3)(7) + (1)(1)}{120}$$

$$P(E) = \frac{63 + 21 + 1}{120} = \frac{85}{120}$$

$P(E) = 0.708$

## Question : 17

**a)**

Data:-

White = 15 , Black = 7

Yellow = 8 , $n(S) = \binom{30}{3}$

**a:**

Let Event: P ( three are Yellow)

S M T W T F S DATE:

$$P(F) = \frac{n(F)}{n(S)} = \frac{\binom{8}{3}\binom{7}{0}\binom{15}{0}}{\binom{30}{3}} = \frac{(56)(1)(1)}{}$$

$$P(F) = \frac{56}{4060} = 0.014$$

**b-**

Let event : G (three ball are of same color)

$$P(G) = \frac{\binom{8}{3}\binom{7}{0}\binom{15}{0} + \binom{8}{0}\binom{7}{3}\binom{15}{0} + \binom{8}{0}\binom{7}{0}\binom{15}{3}}{\binom{30}{3}}$$

$$P(G) = \frac{(56)(1)(1) + (1)(35)(1) + (1)(1)(455)}{4060}$$

$$P(G) = \frac{546}{4060} = 0.134$$

**c-**

Let event H : ( three balls are of different colors)

$$P(H) = \frac{n(S_H)}{n(S)}$$

$$P(H) = \frac{\binom{8}{1}\binom{7}{1}\binom{15}{1}}{\binom{30}{3}} = \frac{(8)(7)(15)}{4060} = \frac{840}{4060}$$

$$P(H) = 0.207$$